

15 September
Monday

Random variable

Discrete RV

- * finite
- * whole numbers
- * eg: dice, coin etc

PMF (probability mass function)
↓
CDF

(Cumulative distribution function)

Continuous RV

- infinite / countable fractions
- eg: range of values within a limit

PDF

(probability density function)

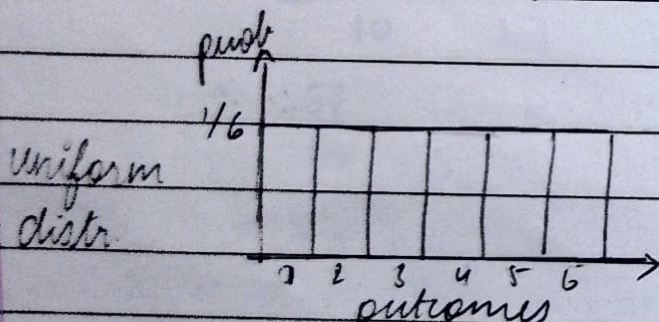
CDF

(cumulative distribution function)

Relationship between PMF, PDF & CDF

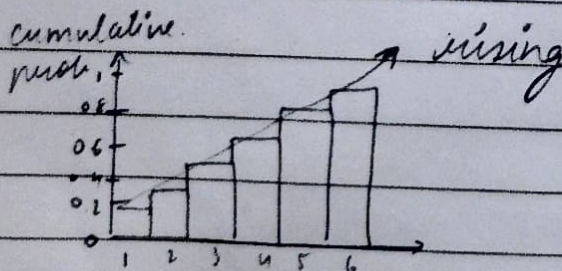
Discrete distribution (PMF)

Dice



equal probability

Cumulative distribution (CDF)



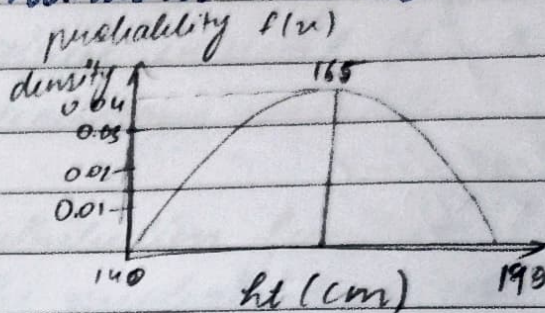
$X = \text{random variable}$

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$$P(X \leq 4) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

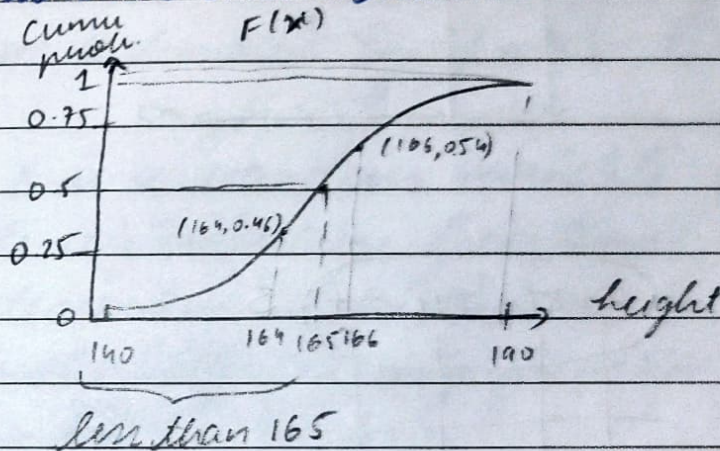
Continuous distribution (PDF)



Gaussian distribution

$$\frac{d}{du} F(x) = f(u)$$

Cumulative distribution



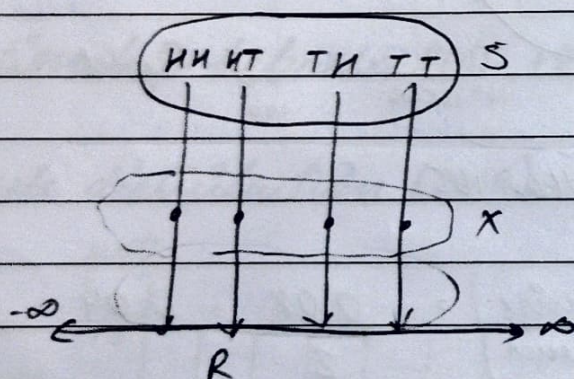
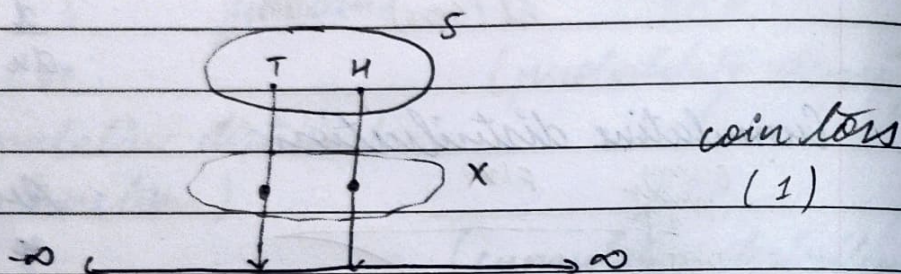
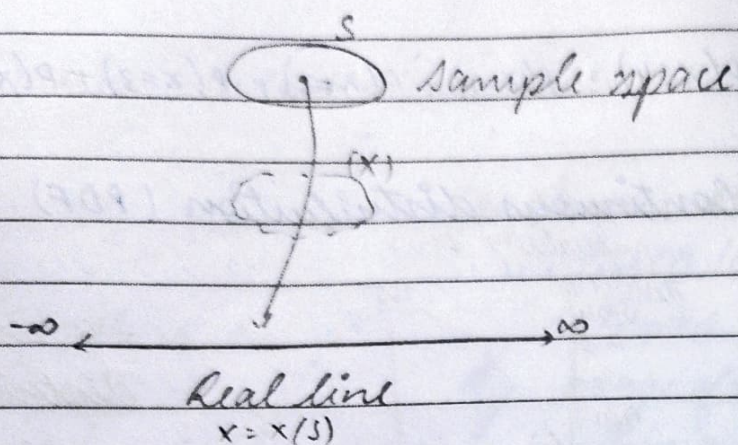
Area ≤ 165

$$\int_{-\infty}^x f(u) du = F(x)$$

$$\text{slope} \left(\frac{\text{rise}}{\text{run}} \right) = \frac{0.08}{2} = 0.04$$

Probability density represents the value of gradient on Cumulative distribution frequency (CDF)

Probability density is equal to the gradient



Discrete random variable

Let X be a random variable. If X takes finite or countably infinite, then X is called discrete random variable.

Continuous random variable

Let X be a random variable, if X takes uncountably infinite values in a given interval then X is called continuous random variable.

Distribution function & Density function

* The probability of the event $\{X \leq u\}$ is called the cumulative distribution function or distribution function of X and is denoted by $F_u(x) = P(X \leq u)$

* If X is a random variable and if $F_u(x)$ is its distribution function, then the derivative of $F_u(x) \in u$ is called probability density function or density function.

$$f_u(x) = \frac{dF_u(x)}{du}$$

Q Let X be a random variable taking values 1, 2 & 3, with probabilities $3/15$, $7/15$, $5/15$. Find its distribution function and show the distribution function diagrammatically.

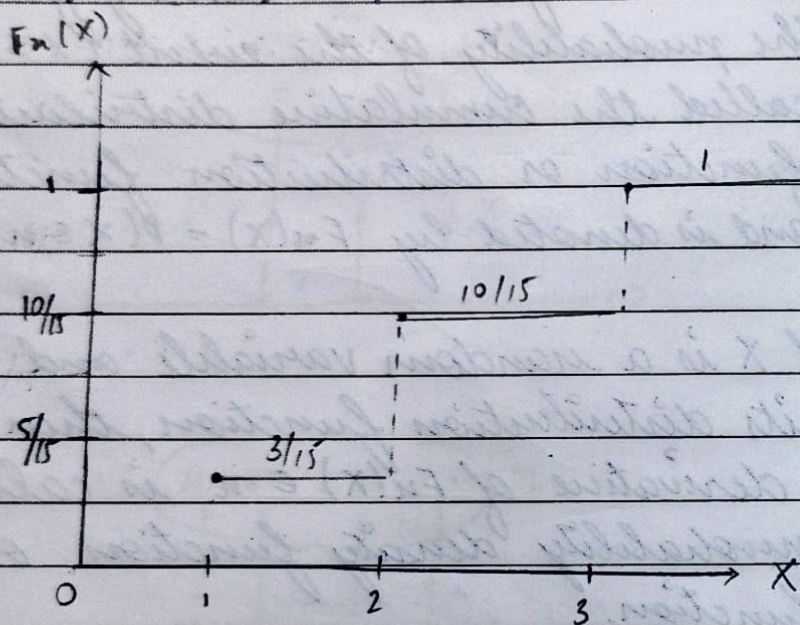
Distribution function

$X:$	1	2	3
$P(X=x):$	$3/15$	$7/15$	$5/15$

$$F_n(X \leq 1) = P(X=1) = 3/15$$

$$F_n(X \leq 2) = P(X \leq 2) = 3/15 + 7/15 = 10/15$$

$$F_n(X \leq 3) = P(X \leq 3) = 3/15 + 7/15 + 5/15 = 1$$



8. Write down the probability density function and also the distribution function of the sum of the numbers appearing when a pair of dice is thrown.

Probability density function

$X:$	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$	$6/36$	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$

	1	1,1	1,2	1,3							6
+	1		2,1	2,2							6
	2		3	3,1							12
				3							

prob. of success (p)

$P(X=n)$ → No. of trial conducted. (n)

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Distribution function

X:	2	3	4	5	6	7	8	9	10	11	12
$P(X=n)$:	$1/36$	$3/36$	$6/36$	$10/36$	$15/36$	$21/36$	$26/36$	$30/36$	$33/36$	$35/36$	$36/36$

Properties of distribution function (NO THEORY)

- * $0 \leq F_n(x) \leq 1$ ($x \leq n$)
- * $F_n(-\infty) = 0$ $-\infty$ {no values exists}
- * $F_n(\infty) = 1$ summation of events = 1
- * $F_n(x_1) \leq F_n(x_2)$ if $x_1 < x_2$ mutually exclusive
- * The function is continuous on right
 $F_n(x^+) = F_n(x)$ CRV
- * $F_n(x_2) - F_n(x_1) = P(x_1 < x \leq x_2)$ DF
- * $F'(n) = f(n)$ If cumulative CRV, PDF is given by

$$\lim_{n \rightarrow 0^+} \frac{P(n \leq X \leq n+h)}{h}$$
- * If random variable X is discrete then $F_n(x)$ is a step function.

Properties of probability density function (NO THEORY)

- * $f_n(x) \geq 0$ for all n {lies above x axis, the curve}
- * $\int_{-\infty}^{\infty} f_n(x) dx = 1$ under a limit, the cumulative area is 1
- * $F_n(x) = \int_{-\infty}^x f_n(t) dt$ answer to distribution function
- * $P(\alpha < x < \beta) = \int_{\alpha}^{\beta} f_n(x) dx$

within a limit of α & β , the area is limited by α & β .

Expectation of random variable

$$E(n) = \sum p_i n_i, \quad \sum p_i = 1 \Rightarrow \text{discrete random var.}$$

$$E(n) = \int_{-\infty}^{\infty} n \cdot f(n) dn \quad \text{where} \quad \int_{-\infty}^{\infty} f(n) dn = 1 \Rightarrow \text{CRV}$$

p_i = probability, n_i = random variable.

- Q A fair coin is tossed 3 times. A person receives rupees X^2 if he gets X heads. Find his expectation.

$$S = \{HHH, HHT, HTH, THH, THT, THT, HTT, TTT\}$$

Probability distribution of X

X :	0	1	2	3
$P(X=n)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

for X^2

X^2 :	0	1	4	9
$P(X^2=n)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$E(X) = \sum p_i n_i$$

$$= \left(\frac{1}{8} \times 0 \right) + \left(\frac{3}{8} \times 1 \right) + \left(\frac{3}{8} \times 4 \right) + \left(\frac{1}{8} \times 9 \right)$$

$$= \frac{24}{8} \text{ rupees}$$

$$= 3 \text{ rupees.}$$

Q Let X denote the order of the Toss at which we

A Fair coin is tossed till a head appears. What is the expectation of the number of Tosses required.

Event	H	TH	TTH	TTTH	...
X	1	2	3	4	
$P(X=n)$	$1/2$	$1/4$	$1/8$	$1/16$	

$$E(X) = \sum p_i x_i$$

$$S = \frac{1 \times 1}{2} + \frac{1 \times 2}{4} + \frac{1 \times 3}{8} + \frac{1 \times 4}{16} + \dots$$

$$\frac{1}{2} S = \frac{1}{4} + \frac{2 \times 1}{8} + \frac{3 \times 1}{16} + \dots$$

$$S - \frac{1}{2} S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$= \frac{1}{2} \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right] \Rightarrow \text{GP series } \left[\frac{a}{1-r} \right]$$

$$\frac{1}{2} S = \frac{1}{2} \left[\frac{1}{1 - 1/2} \right] = 1$$

$$S = E(X) = 1 + 1 = 2$$

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$E(X)$

Q Find expectation of X if X has the PDF

$$f(u) = \begin{cases} ku(2-u), & 0 \leq u \leq 2, k > 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$E(X) = \int_0^2 u f(u) du$$

$$= k \int_0^2 (2u^2 - u^3) du$$

$$= \int_0^2 u \cdot ku(2-u)$$

$$= k \int_0^2 2u^2 du - k \int_0^2 u^3 du$$

$$= 2k \frac{u^3}{3} - k \frac{u^4}{4}$$

$$= 2k \left(\frac{8}{3} \right) - k \left(\frac{16}{4} \right)$$

$$= \frac{16k}{3} - 4k$$

$$= \frac{4k}{3}$$

Q The daily consumption of electric power in million KWh is a random variable X with probability density function.

$$f(u) = \begin{cases} k u e^{-u/3} & \text{for } u > 0 \\ 0 & \text{for } u \leq 0 \end{cases}$$

Find the value of k , expectation and the probability that on a given day the electric consumption is more than expected value.

* For k

$$\int_{-\infty}^{\infty} f(u) du = 1$$

$$k \int_0^{\infty} u e^{-u/3} du = 1$$

$$k \left[u \int_0^{\infty} e^{-u/3} du - e^{-u/3} \int_0^{\infty} u du \right] = 1$$

$$k \left[u \left(-3 e^{-u/3} \right) - e^{-u/3} \frac{u^2}{2} \right] = 1$$

k

$$k \left[u \int_0^{\infty} e^{-u/3} du - \int_0^{\infty} 1 \times \left(-3 e^{-u/3} \right) du \right]$$

$$k \left[u \left(-3 e^{-u/3} \right) + 3 \left(-3 e^{-u/3} \right) \right] = 1$$

$$k \left[-u3(1) + 9 \right] = 1$$

$$k = \frac{1}{9}$$

$$= \frac{1}{9} \int_0^{\infty} u^2 e^{-u/3} du$$

$$K \left[u \left(-\frac{3}{e^{3/u}} \right) \right] \rightarrow \frac{9}{e^{3/u}} = 1$$

Applying the limits

$$K \left[-3(\infty) e^{-3/\infty} \right] - 9e^{-3/0} - \left[-3(0) e^{-3/0} \right] - 9e^{3/0} = 1$$

$$K \cdot 9 = 1$$

$$K = \frac{1}{9}$$

$$* E(X) = \int_{-\infty}^{\infty} u f(u) du = \frac{1}{9} \int_0^{\infty} u^2 e^{-u/3} du \quad \text{gamma}$$

$$= \text{where } \alpha = 3; \lambda = 1/3$$

$$= \text{Using gamma function}$$

$$= \frac{\Gamma(\alpha)}{\lambda^\alpha}$$

$$= \frac{2\sqrt{3}\sqrt{3}}{1}$$

$$= \left(\frac{5}{3}\right)^3$$

$$= 27\sqrt{3}$$

$$= \frac{27 \times 2}{9}$$

$$= 6$$

$$= 6$$

Moments { 1 order = mean
2 order = variance

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$$* P(X > 6) = \frac{1}{9} \int_6^{\infty} f(u) du$$

$$= \frac{1}{9} \int_6^{\infty} u e^{-u/3} du$$

$$= \frac{1}{9} \left[u(-3e^{-u/3}) - \int (-3e^{-u/3}) du \right]_6^{\infty}$$

$$= \frac{1}{9} \left[-3ue^{-u/3} - 9e^{-u/3} \right]_6^{\infty}$$

0.406

$$= \frac{1}{9} \left[-3(0)(0) - 9(0) \right] - \left[-3(6)(e^{-2}) - 9(e^{-2}) \right]$$

$$= \frac{1}{9} \left[0 + 9e^{-2} + 9e^{-2} \right] = \frac{2}{e^2} \approx 0.406$$

Mean

$$\mu'_1 = E(X) = \sum p_i x_i \quad \text{or} \quad \mu'_1 = \int_{-\infty}^{\infty} u f(u) du$$

$$\mu'_2 = E(X^2) = \sum p_i x_i^2 \quad \text{or} \quad \mu'_2 = \int_{-\infty}^{\infty} u^2 f(u) du$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \mu'_2 - (\mu'_1)^2$$

discrete

Q. If X denotes the smaller of the 2 numbers including equality that appears when a pair of dice is thrown. Find the probability distribution of X also find mean and variance of X .

Probability distribution of X

$X:$	1	2	3	4	5	6
$P(X=x):$	$1/36$	$9/36$	$7/36$	$5/36$	$3/36$	$1/36$

$$E(X) = \left(1 \times \frac{1}{36}\right) + \left(2 \times \frac{9}{36}\right) + \left(3 \times \frac{7}{36}\right) + \left(4 \times \frac{5}{36}\right) + \left(5 \times \frac{3}{36}\right) + \left(6 \times \frac{1}{36}\right)$$

$$= \frac{91}{36}$$

$$= 2.52\bar{7}$$

$$E(X^2) = \frac{301}{36}$$

$$= 8.36\bar{1}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$= \underline{1.975}$$

Q. A discrete random variable has the probability density function given below

$x:$	-2	-1	0	1	2	3
$P(X=x):$	0.2	K	0.1	$2K$	0.1	$2K$

$$\sum p_i = 1 \Rightarrow 0.2 + K + 0.1 + 2K + 0.1 + 2K = 1$$

$$5K + 0.4 = 1$$

$$5K = 0.6$$

$$K = 0.12$$

X	-2	-1	0	1	2	3
$P(X=x)$	0.2	0.12	0.1	0.24	0.1	0.24
$E(X) =$	0.64					
$E(X^2) =$	3.72					

$$\begin{aligned}
 V(X) &= E(X^2) - [E(X)]^2 \\
 &= 3.72 - 0.4096 \\
 &= \boxed{3.3104}
 \end{aligned}$$

Continuous

- Q. A continuous random variable X has the PDF defined by $f(u) = A + Bu$, $0 \leq u \leq 1$. If the mean of distribution is $\frac{1}{3}$, find A and B .

$$\int_{-\infty}^{\infty} f(u) du = 1$$

$$\int_0^1 (A + Bu) du = 1$$

$$E(u) = \int_0^1 u f(u) du$$

$$\int_0^1 u(A + Bu) du = \frac{1}{3}$$

$$\left[\frac{Au}{2} + \frac{Bu^2}{2} \right]_0^1 = 1$$

$$(A + B) - (0) = 1$$

$$A + B = 1$$

 $\Rightarrow$

$$\boxed{2A + B = 2}$$

$$\left[\frac{Au^2}{2} + \frac{Bu^3}{3} \right]_0^1 = \frac{1}{3}$$

 $\frac{1}{3}$

$$\frac{A}{2} + \frac{B}{3} = \frac{1}{3}$$

 $\Rightarrow$

$$\boxed{3A + 2B = 2}$$

$$A = 2 \quad \& \quad B = -2$$

$$\text{PDF: } f(u) = 2 - 2u, \quad 0 \leq u \leq 1$$

Q A continuous random X has the PDF $f(u) = ku^2e^{-u}$, $u \geq 0$
find K, mean and variance.

$$\int_{-\infty}^{\infty} f(u) = 1$$

$$\int_0^{\infty} ku^2e^{-u} = 1$$

$$k \left[u^2(-e^{-u}) - \int 2u(-e^{-u}) \right] du = 1$$

$$k \left[-u^2e^{-u} - 2 \int u(e^{-u}) - \int \{e^{-u}\} du \right] \Big|_0^{\infty} = 1$$

$$k \left[-u^2e^{-u} - 2 \left[-ue^{-u} - e^{-u} \right] \right] \Big|_0^{\infty} = 1$$

$$k \left[-u^2e^{-u} + 2ue^{-u} + 2e^{-u} \right] \Big|_0^{\infty} = 1$$

$$k \left[-(0)^2e^{-0} + 2(0)e^{-0} + 2(e^{-0}) \right] = 1$$

$$k [2] = 1$$

$$k = \frac{1}{2}$$

$$\begin{aligned}
 E(X) &= \frac{1}{2} \int_0^{\infty} u^3 e^{-u} du \\
 &= \frac{1}{2} \left[u^3 (-e^{-u}) - \int (3u^2 \times (-e^{-u})) du \right]_0^{\infty} \\
 &= \frac{1}{2} \left[-u^3 e^{-u} - \int 3u^2 (e^{-u}) - \int (6u \times e^{-u}) du \right]_0^{\infty} \\
 &= \frac{1}{2} \left[-u^3 e^{-u} - \int 3u^2 e^{-u} - \int 6u (-e^{-u}) - \int (6(-e^{-u})) \right]_0^{\infty} \\
 &= \frac{1}{2} \left[-u^3 e^{-u} - \int 3u^2 e^{-u} - \int -6u e^{-u} - 6e^{-u} \right]_0^{\infty} \\
 &= \frac{1}{2} \left[-u^3 e^{-u} - 3u^2 e^{-u} + 6u e^{-u} + 6e^{-u} \right]_0^{\infty} \Rightarrow \boxed{\frac{6}{2}} \\
 &= 3
 \end{aligned}$$

$$\mu_1' = E(X) = \frac{1}{2} \int_0^{\infty} u^3 e^{-u} du = 3$$

$$\mu_2' = \frac{1}{2} \int_0^{\infty} u^2 f(u) du = \frac{1}{2} \int_0^{\infty} u^4 e^{-u} du = 12$$

$$V(u) = \mu_2' - (\mu_1')^2 = 12 - 3^2 = 3$$

Q A continuous random variable has the following probability law
 $f(x) = kx^2, 0 \leq x \leq 2$
 Find the normalizing factor k and find the probabilities.

- ① $k?$ ② $P(0.2 \leq u \leq 0.5)$ ③ $u^{3/4}$ given that $u \geq 1/2$

④ Obtain distribution function

$$1. \int_{a_1}^b f(u) du = 1$$

$$\int_0^1 k u^2 du = 1$$

$$k \left[\frac{u^3}{3} \right]_0^1 = 1$$

$$k \frac{8}{3} = 1$$

$$k = \frac{3}{8}$$

$$2. P(0.2 \leq u \leq 0.5)$$

$$\frac{3}{8} \int_{0.2}^{0.5} u^2 du =$$

only $f(u)$:
no u part

$$\frac{3}{8} \left[\frac{u^3}{3} \right]_{0.2}^{0.5} = 0.014625$$

$$3. P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A) = P(X \geq 1/2) = 0.9843$$

$$P(B) = P(X \geq 3/4) = 0.947$$

$$P(A \cap B) = 0.947$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow \frac{3}{8} \int_{0.5}^2 u^2 du$$

$$\frac{3}{8} \left[\frac{u^3}{3} \right]_{0.5}^2 = 0.9843$$

$$\Rightarrow \frac{3}{8} \int_{0.75}^2 u^2 du = 0.947 \quad P(A \cap B)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.947}{0.9843} = 0.9621$$

$$4. \quad F_u(x) = \int_{-\infty}^x f(u) du = \int_{-\infty}^x \frac{3}{8} u^2 du = \frac{u^3}{8}$$

$$= \frac{3}{8} \int_0^x u^2 du \quad \left[\frac{3 \times u^3}{8 \times 3} \right]$$

$$= \frac{3}{8} \left[\frac{u^3}{3} \right]_0^x$$

$$= \frac{u^3}{8}$$

a Let X be a continuous random variable with PDF

$$f_u(x) = ku(1-u), \quad 0 \leq u \leq 1$$

Find k , and determine number 'b' such that probability $P(X \leq b) = P(X \geq b)$ also obtain distribution function.

$$1. \int_a^b f(u) du = 1$$

$$\int_0^1 ku(1-u) du = 1$$

$$k \left[\frac{u^2}{2} - \frac{u^3}{3} \right]_0^1 = 1$$

$$k \left[\frac{1}{2} - \frac{1}{3} \right] = 1$$

$$k \left[\frac{1}{6} \right] = 1$$

$$k = 6$$

$$2. P(X \leq b) = P(X \geq b)$$

Total probability = 1

$$P(X \leq b) = 1/2$$

$$\int_0^b f(u) du = 1/2$$

$$6 \int_0^b (u - u^2) du = 1/2$$

$$6 \left[\frac{u^2}{2} - \frac{u^3}{3} \right]_0^b$$

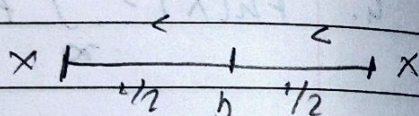
$$6 \left[\frac{b^2}{2} - \frac{b^3}{3} \right] = \frac{1}{2}$$

$$3b^2 - 2b^3 = \frac{1}{2}$$

$$6b^2 - 4b^3 = 1$$

$$4b^3 - 6b^2 + 1 = 0$$

$$b = 0.5$$



$$-0.36 \times$$

$$b = 1.36 \times$$

$$0.5 \checkmark$$

$$\begin{aligned}
 3. \quad F_u(x) &= \int_{-\infty}^u f(u) du = \int_0^u (u - u^2) du \\
 &= 3u^2 - 2u^3; 0 \leq u \leq 1
 \end{aligned}$$

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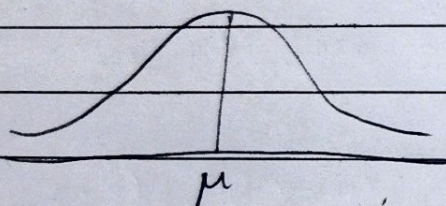
Thursday

Types of Probability Distribution

Discrete
Bernoulli
Binomial
Uniform
Poisson

Continuous
Normal
Chi-squared
Exponential
Logistic $\propto$
Student T $\propto$

Normal distribution



mean, median or mode

Binomial distribution (2 unknown: n & p)

Measures of central tendency & dispersion
 $\mu = np$ (mean)

where

μ = mean of binomial distribution

n = number of trials

p = probability of success

$$\sigma = \sqrt{npq} \quad (\text{SD})$$

where

σ = standard deviation

q = probability of failure ($1-p$)

~~nC_r~~

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

probability $\hookrightarrow$ number of trials.

Assumptions

1. No. of experiment conducted are finite 'n'
2. Probability of success is 'p' & constant for each exp. * n to value is very low compared to p

Q A coin is biased so that head is twice as likely to appear as a tail. If the coin is tossed 6 times, find the probability of getting

1. Exactly 2 heads
2. Atleast 3 heads
3. Atmost 3 heads

$$H = 2T$$

$$3T = 1$$

$$T = \frac{1}{3}, H = \frac{2}{3}$$

$$p = \frac{2}{3}, q = \frac{1}{3}$$

$$\begin{aligned} 1. P(X=2) &= {}^nC_r \cdot p^r \cdot q^{n-r} \\ &= {}^6C_2 \left(\frac{2}{3}\right)^2 \cdot \left(\frac{1}{3}\right)^4 \\ &= 0.082 \end{aligned}$$

$$2. P(X \geq 3) = P(X=3) + P(X=4) + P(X=5) + P(X=6)$$

or

$$1 - P(X < 3)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - [0.0003 + 0.0164 + 0.0823]$$

$$\approx 0.9 \approx 0.899245$$

3. $0.1 + P(X=3)$
 $P(X < 3) + P(X=3)$
 $0.1 + 0.2194$
 $= 0.3194$

Q A binomial distribution with parameter $n=5$ satisfy the property $8P(X=4) = P(X=2)$

1. Find the value of P

2. $P(X \geq 1)$

$$8P(X=4) = \frac{2}{5}P(X=2)$$

$$P(X=4) = \frac{P(X=2)}{8}$$

$$P(X=1) + 8P(X=4)$$

$$P(X=2) = {}^5C_2 \times p^2 \times q^3$$

$$P(X=4) = {}^5C_4 \times p^4 \times q$$

$$8 \times {}^5C_4 \times p^4 \times q = {}^5C_2 \times p^2 \times q^3$$

$$\frac{p^2}{q^2} = \frac{{}^5C_2}{{}^5C_4}$$

$$q^2 = 8C4$$

$$p^2 = q^2 \cdot 0.25$$

$$\frac{p}{q} = 0.5$$

$$p = q \cdot 0.5$$

$$p = (1-p) \cdot 0.5$$

$$p + 0.5p = 0.5$$

$$1.5p = 0.5$$

$$p = \frac{0.5}{1.5} = \frac{1}{3} ; q = \frac{2}{3}$$

random variable

binomial dist

Poisson dist + geometric distribution

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$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - (P(X=0) + \cancel{P(X=1)})$$

$$= 1 - (0.1316 + \cancel{0.3292})$$

$$= \cancel{0.5392} \quad 0.868$$

- Q. The mean and variance of binomial variable are 12 & 4. Find the distribution

$$\mu = np$$

$$12 = np$$

$$n = \frac{6}{2} \times 3$$

$$\sigma = \sqrt{npq}$$

$$2 \quad 4 = \sqrt{npq}$$

$$2 \quad 16 = npq$$

$$\frac{16}{12} = q$$

$$\frac{4}{3} = q$$

$$\frac{4}{3} = q$$

$$p = \frac{2}{3}$$

Poisson distribution (1 unknown = d)

* Limiting case of binomial distribution

Assumptions:

- n = number of trial is indefinitely large
 $n \rightarrow \infty$ (tends)

- p = of each trial tends to 0

$p \rightarrow 0$ (tends).

$$P(X=u) = \frac{e^{-\lambda} \cdot \lambda^u}{u!}, u=0, 1, 2, 3, \dots; \lambda=np$$

If 5% of the electric bulbs manufactured by a company are defective. Find the probability that in a sample of 100 bulbs

1. none is defective
2. 5 bulbs are defective

$$\lambda = 5$$

$$1. P(X=0) = \frac{e^{-\lambda} \cdot \lambda^0}{0!} = 0.0067$$

$$2. P(X=5) = \frac{e^{-\lambda} \cdot \lambda^5}{5!} = \frac{e^{-5} \times 5^5}{5!} = 0.175$$

A manufacturer of blades knows that 5% of his product is defective. If he sells the blades in a box of 100 and guarantees that not more than 10 blades are defective. What is the probability that a lion will fail to meet the guaranteed quality.

$$\lambda = np$$

$$= 5$$

$$P(X \geq 10) = 1 - P(X \leq 10) = 0.01$$

$$\begin{aligned}
 &\Rightarrow 1 - \{P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + \\
 &\quad P(X=5) + P(X=6) + P(X=7) + P(X=8) + P(X=9) + P(X=10)\} \\
 &= 1 - \{ (6.734 \times 10^{-8}) + (0.0336) + (0.0842) + (0.1403) + \\
 &\quad (0.1754) + (0.1754) + (0.1462) + (0.1044) + (0.0652) \\
 &\quad + (0.0362) + (0.0181) \} \\
 &= 1 - (0.985737) \\
 &= 0.014263 \approx 0.0137
 \end{aligned}$$

Fit the Poisson distribution to the following data and calculate the theoretical frequencies

x	f	fx	Theoretical
0	123	0	121
1	59	59	81
2	14	28	15
3	3	9	3
4	1	4	0.5
	<u>200</u>	<u>100</u>	<u>200</u>

Theoretical frequency : $\sum f \times \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$$= e^{-0.5} \cdot (0.5)$$

$\lambda = 200$

(until successful event is achieved)
Geometric distribution (unknown = p)
 Number of trials required to get
 first success in a series of independent
 and identically distributed
 Bernoulli trials, where are two
 possibility in outcomes: success &
 failure.

Binomial
 number of trial
 (finite)

↓

n (success) only

Geometric
 number of trial
 (infinite)

↓

1st success at n^{th} point
 failure = $(n-1)$

PMF

$$P(X=n) = (1-p)^{n-1} (p)$$

CDF

$$P(X \leq n) = 1 - (1-p)^n$$

Mean

$$E(X) = \frac{1}{p}$$

Variance

$$\text{Var}(X) = \frac{(1-p)}{p^2}$$

Standard deviation

$$\sigma = \frac{\sqrt{1-p}}{p}$$

where $q = (1-p)$

Suppose that the trainee soldier shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7

- 1) What is the probability that the target would be hit at 10th attempt
- 2) Probability that it takes him less than 4 shots.
- 3) Probability that it takes him an even number of shots.

1. $P(X=10)$

HW

000

$$p = 0.7, \quad q = 0.3$$

$$= q^{n-1} \times p = (0.3)^9 \times 0.7$$

$$= 1.377 \times 10^{-5}$$

2. $P(X < 4) = P(X=1) + P(X=2) + P(X=3)$

$$= p + qp + q^2p$$

$$= 0.973$$

3. $P(X = \text{even}) = P(X=2) + P(X=4) + P(X=6) + P(X=8) + \dots$

$$= qp + q^3p + q^5p + \dots$$

$$= qp [1 + q^2 + q^4 + \dots]$$

$$= qp \left[\frac{1}{1 - q^2} \right]$$

$$= (0.3)(0.7) \left[\frac{1}{1 - (0.3)^2} \right]$$

$$= 0.23$$

4. $P(X = \text{odd}) =$

1. Suppose that the probability for an applicant for a drivers license to pass a road test on any given attempt is $\frac{2}{3}$ (0.66). What is the probability that the applicant will pass the road test on the 3rd attempt.

2. Let X be the number of birth in a hospital until the first girl is born. Determine the probability and distribution function of X . Assume that the probability that the baby born is a girl is 0.5.

1. $P(X=3)$

$$p = 0.667$$

$$= 2/3$$

$$q = 0.333$$

$$= 1/3$$

$$= q^{n-1} \times p$$

$$= q^2 \times p$$

$$= \left(\frac{1}{3}\right)^2 \times \left(\frac{2}{3}\right)$$

$$= 0.074$$

2. $P(X=n)$

$$p = 0.5$$

$$q = 0.5$$

$$PBF = (0.5)^{n-1} (0.5)$$

$$= (0.5)^n$$

$$CDF = 1 - (0.5)^n$$

in M2.